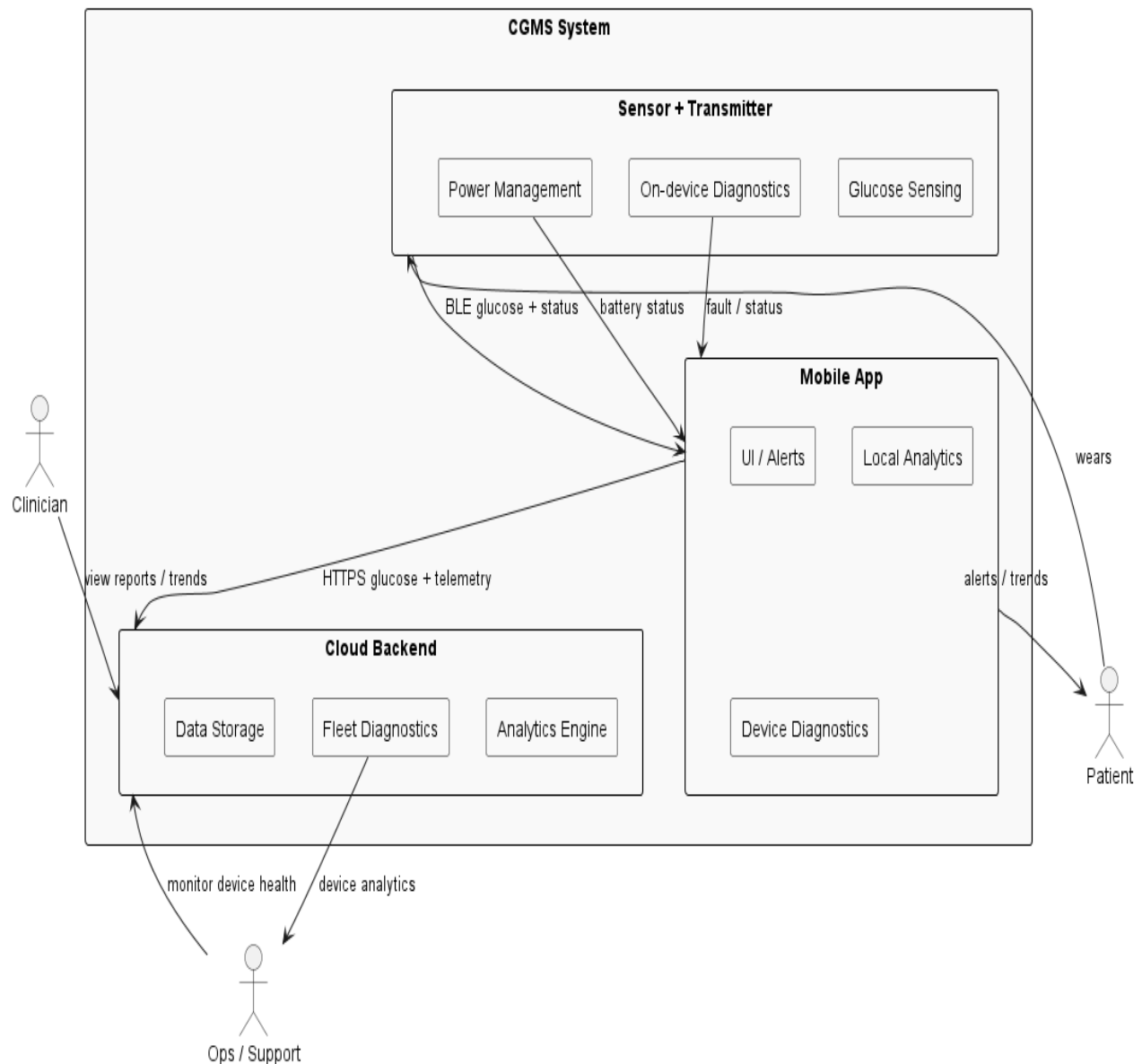


Continuous Glucose Monitoring System (CGMS)

System Architecture & Safety Design

1. System Context

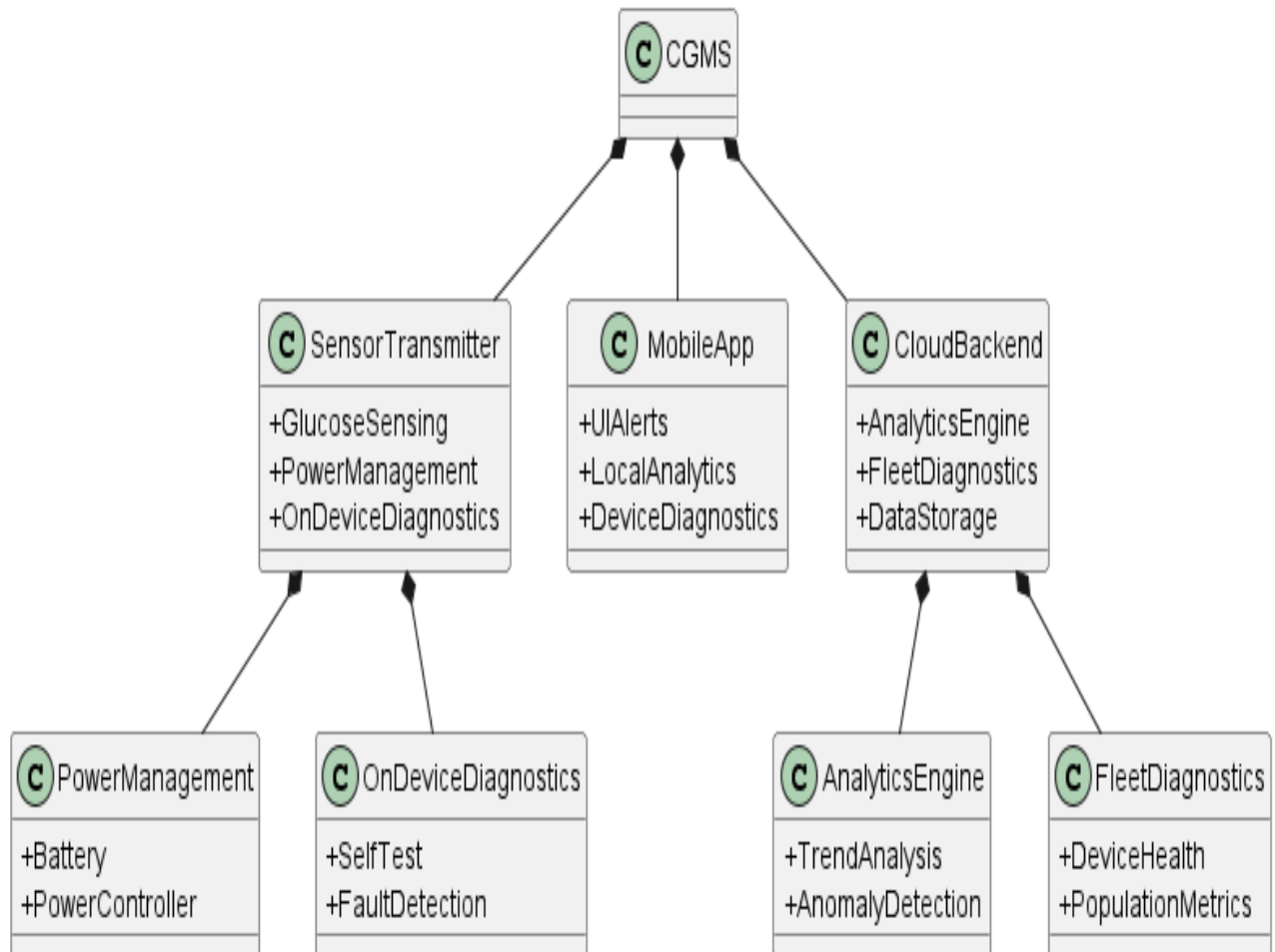


System Context

2. System Architecture (SysML)

2.1 Block Definition Diagram

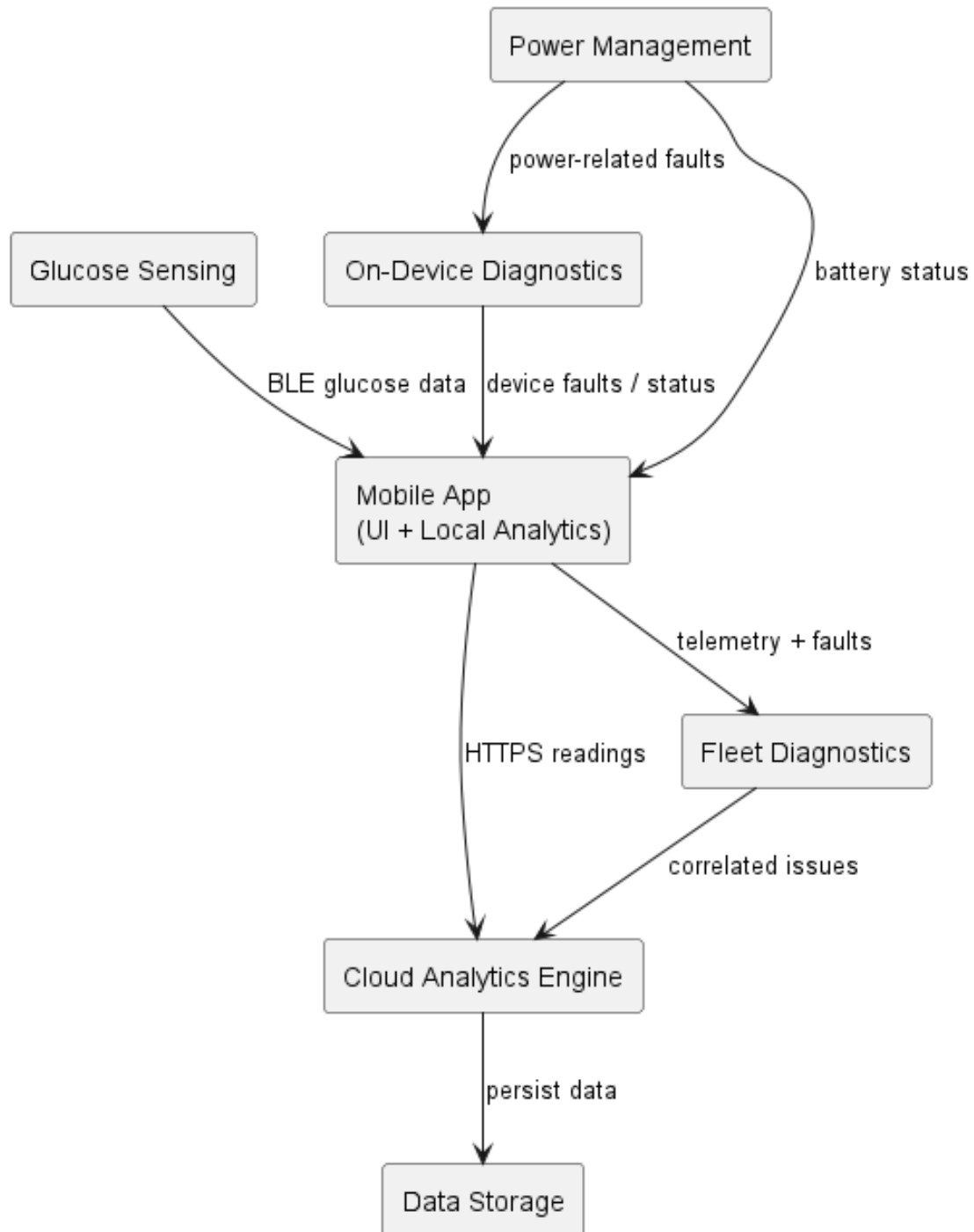
CGMS - Block Definition Diagram (BDD)



BDD

2.2 Internal Block Diagram

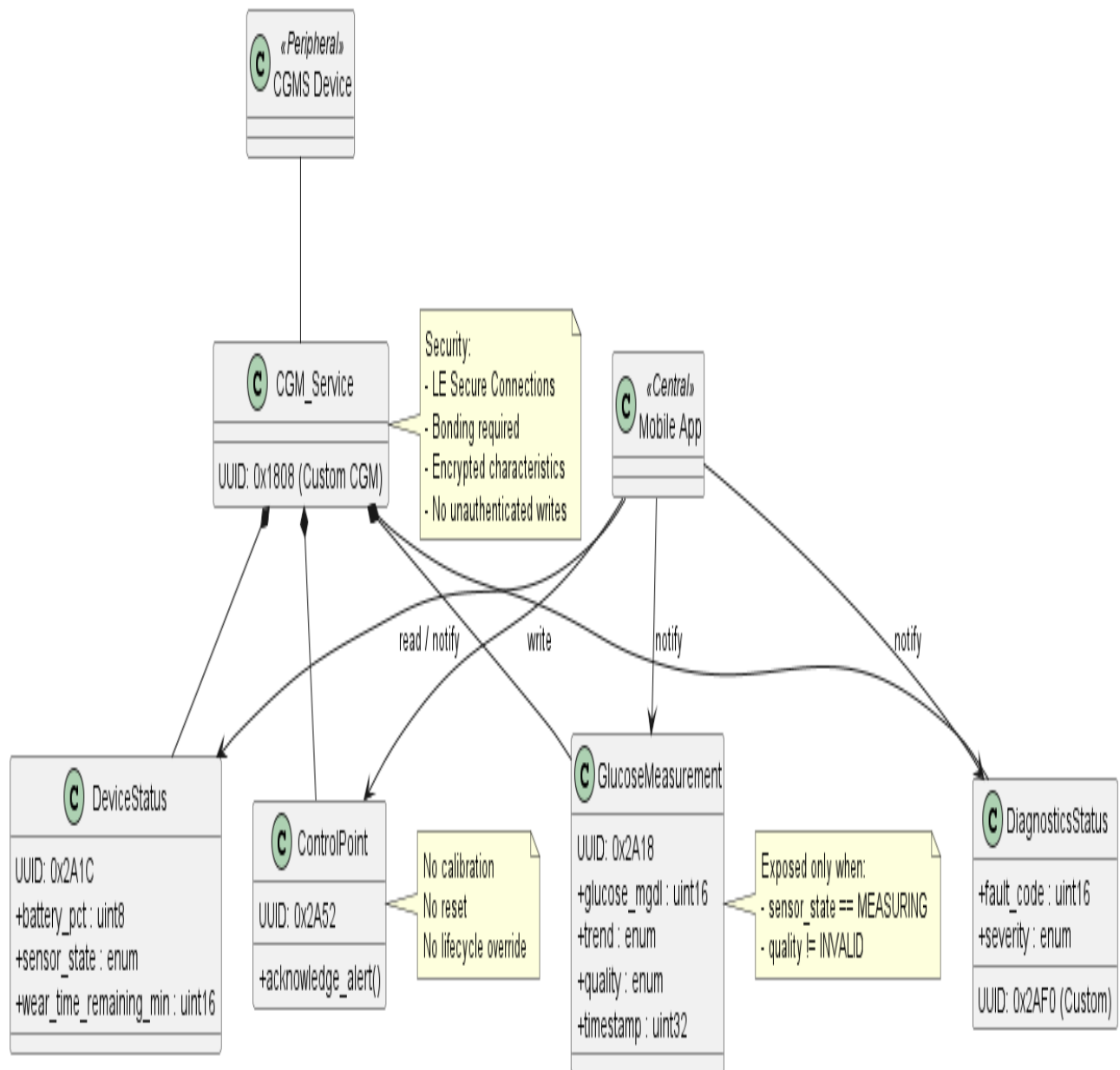
CGMS - Internal Block Diagram (IBD)



3. Interfaces

3.1 BLE GATT Interface

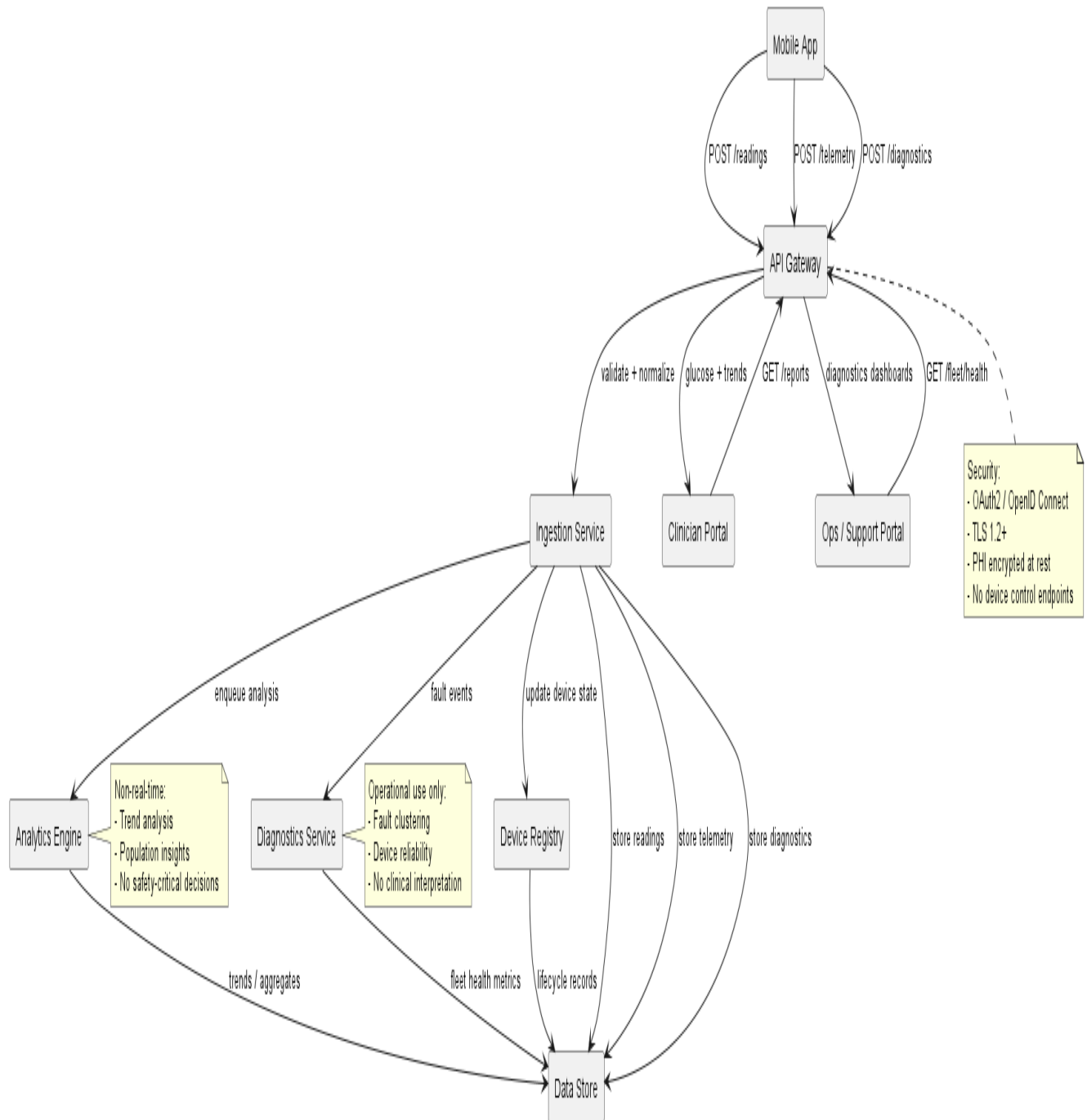
CGMS BLE GATT Profile (Single-Use Sensor)



BLE GATT

3.2 Cloud API Interface

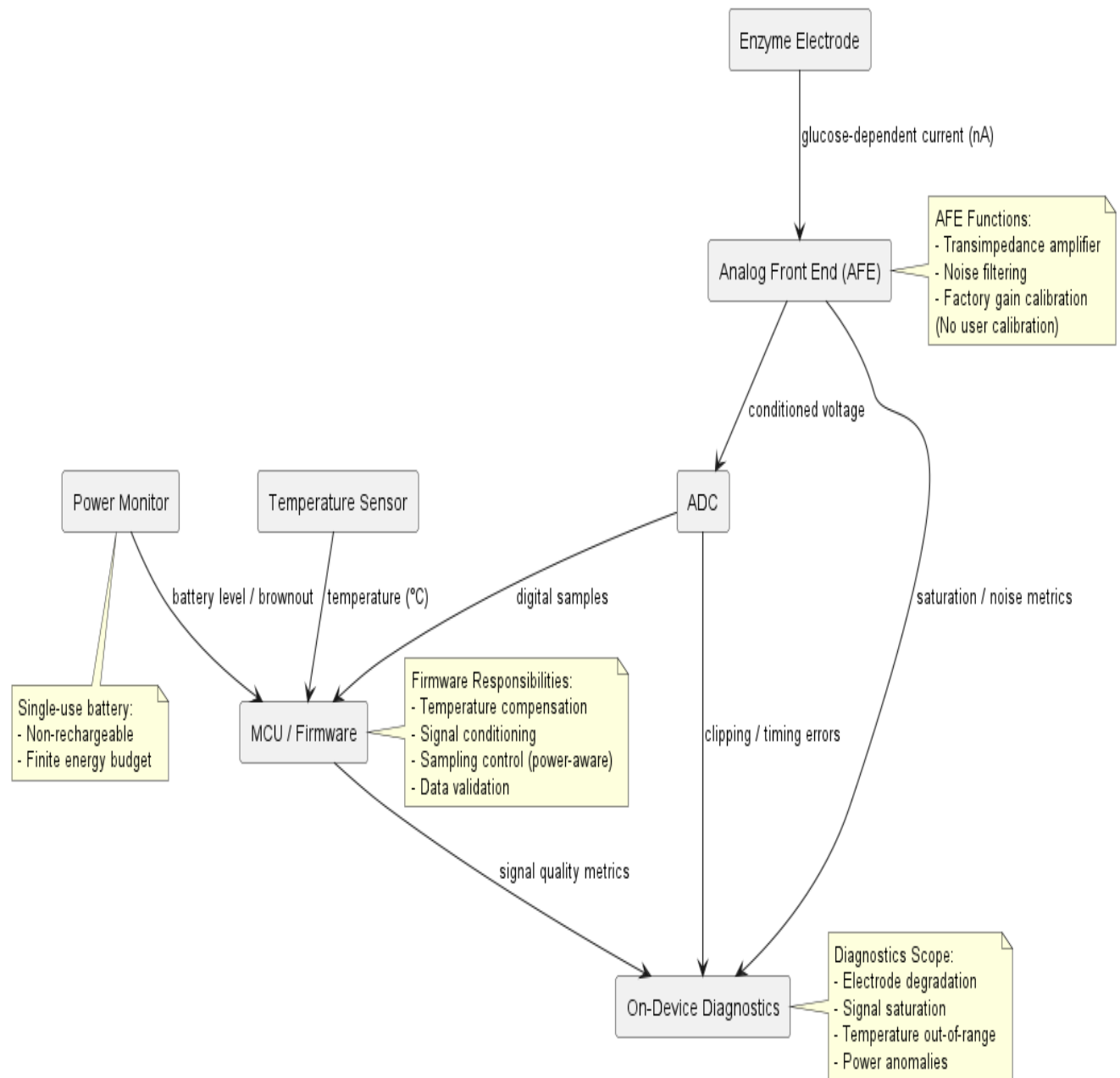
CGMS Cloud API Interface (Updated)



Cloud API

3.3 Sensor Interface

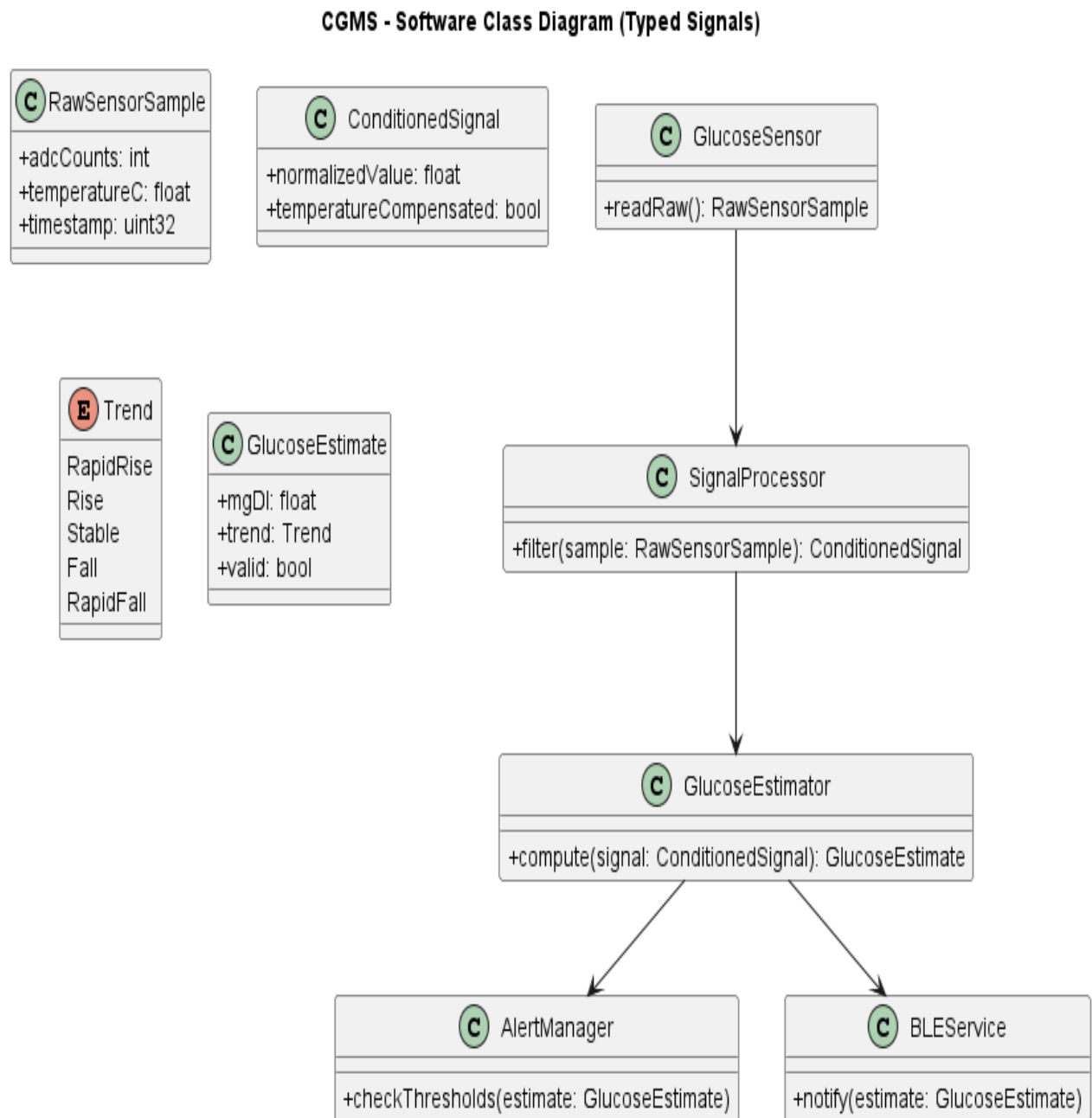
CGMS Sensor Interface (Updated)



Sensor IF

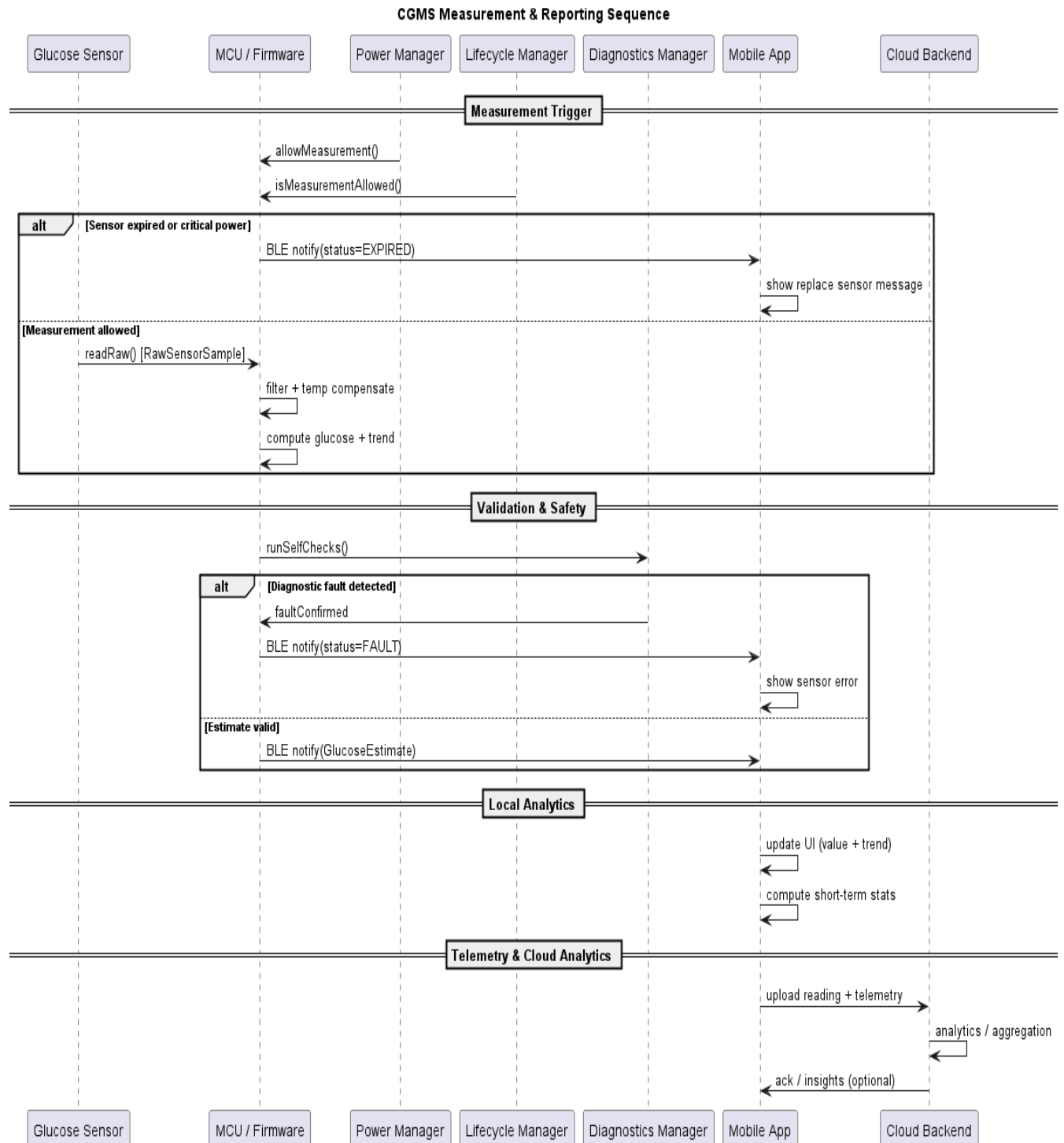
4. Software Design (UML)

4.1 Class Diagram



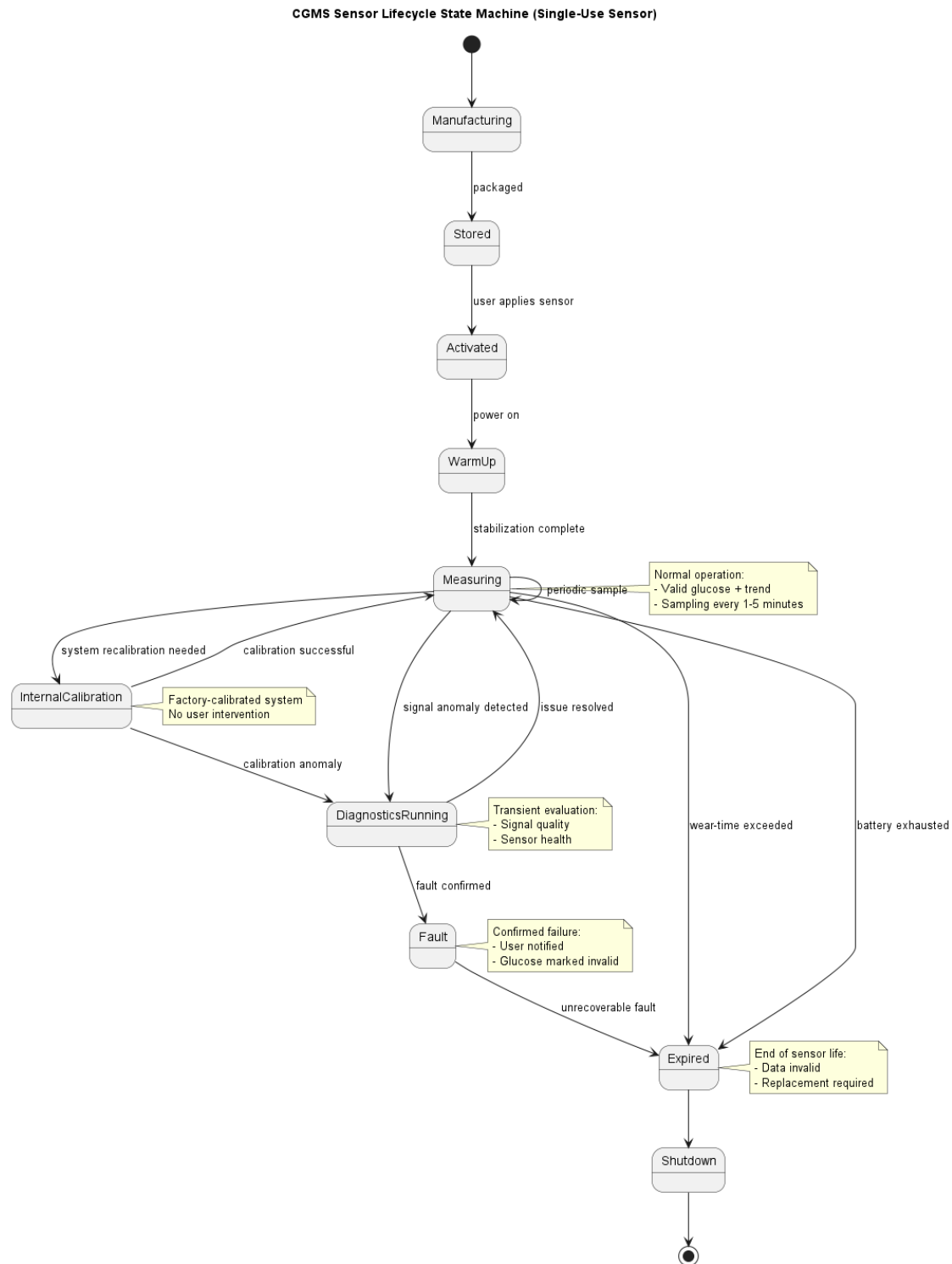
Class Diagram

4.2 Sequence Diagram



Sequence Diagram

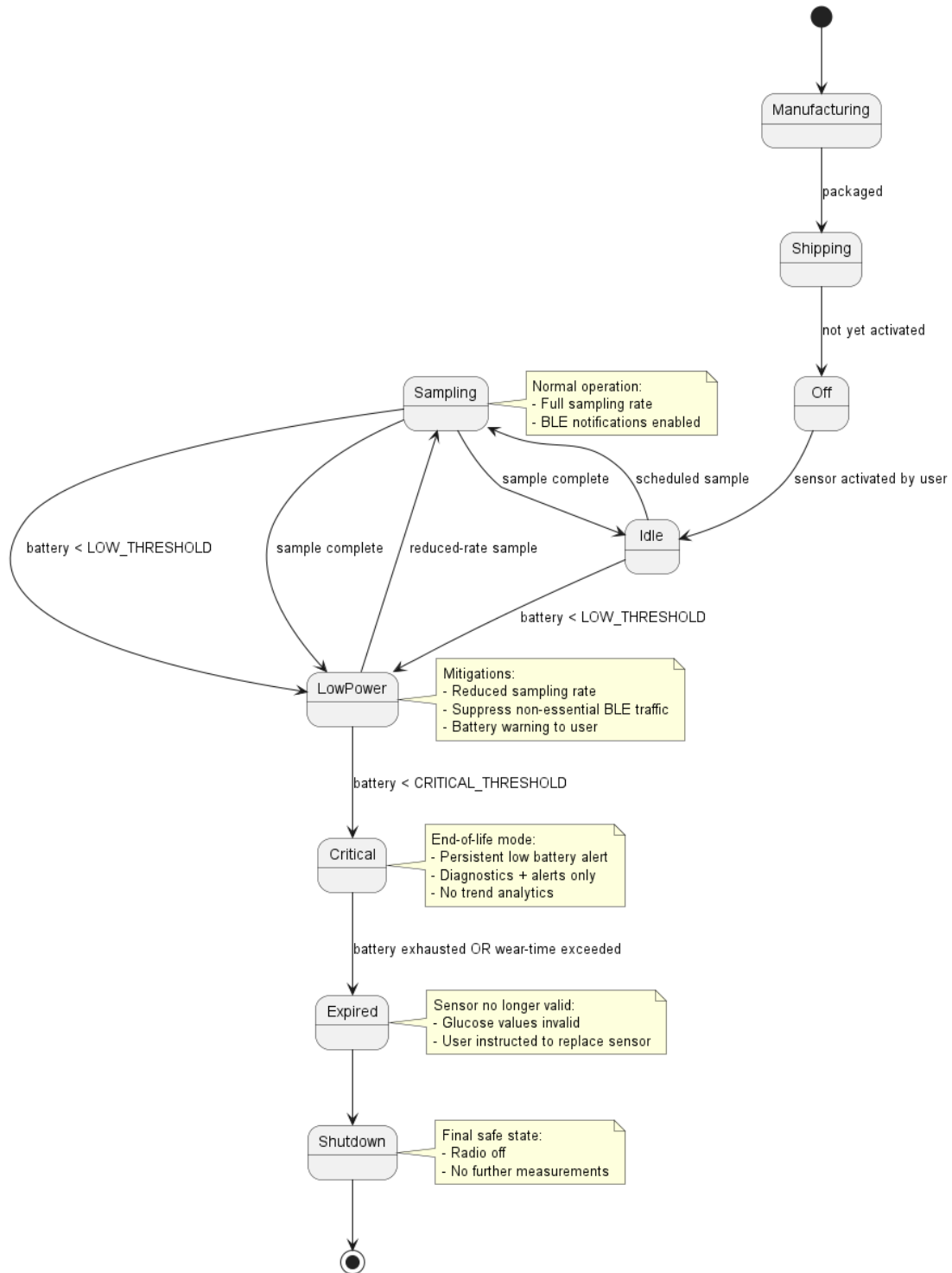
4.3 Sensor Lifecycle State Machine



Lifecycle State Machine

4.4 Sensor Power State Machine

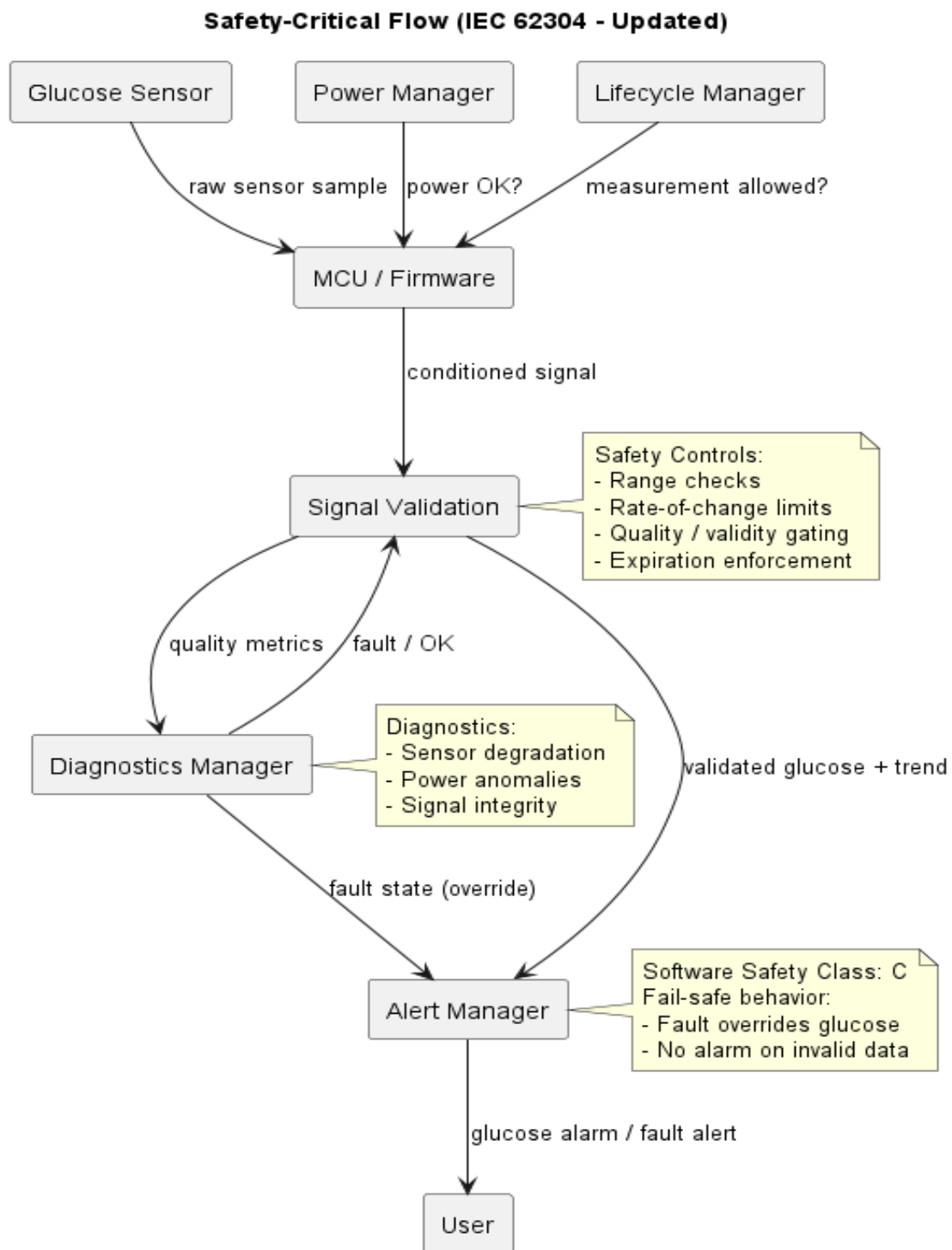
CGMS Power Management - Single-Use Sensor (Non-Rechargeable)



Power State Machine

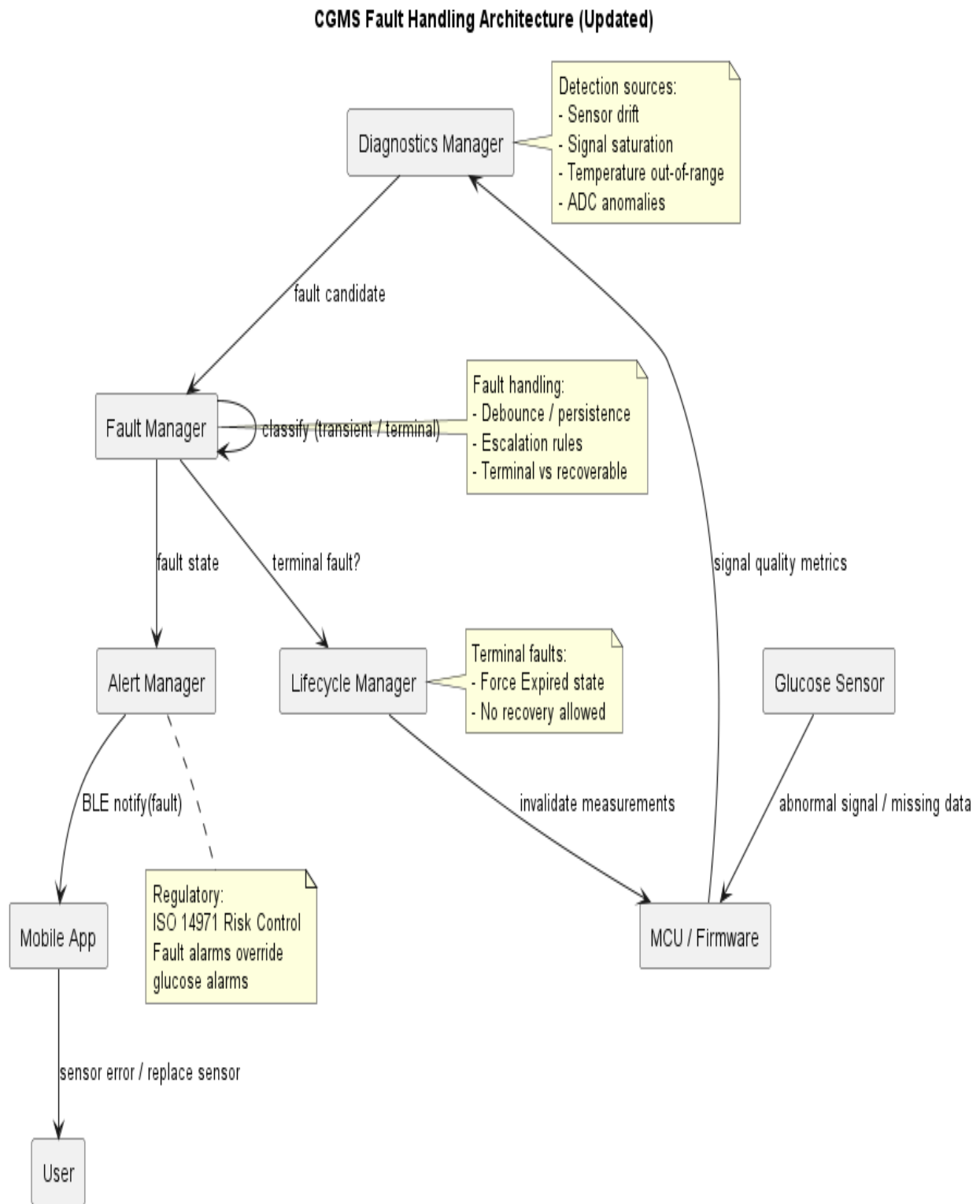
5. Safety & Risk Management

5.1 Safety-Critical Data Flow (IEC 62304)



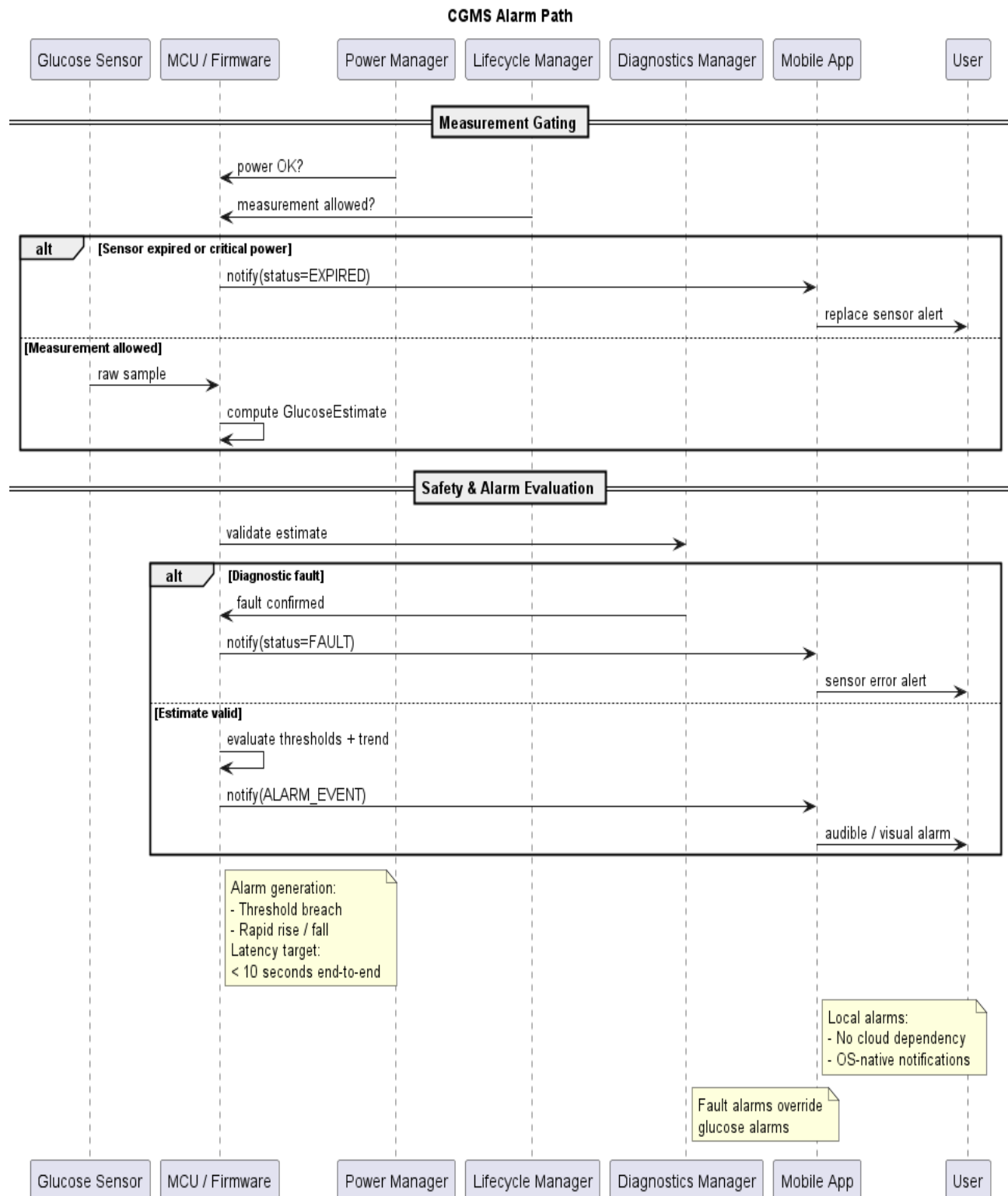
Safety Flow

5.2 Fault Handling



Fault Handling

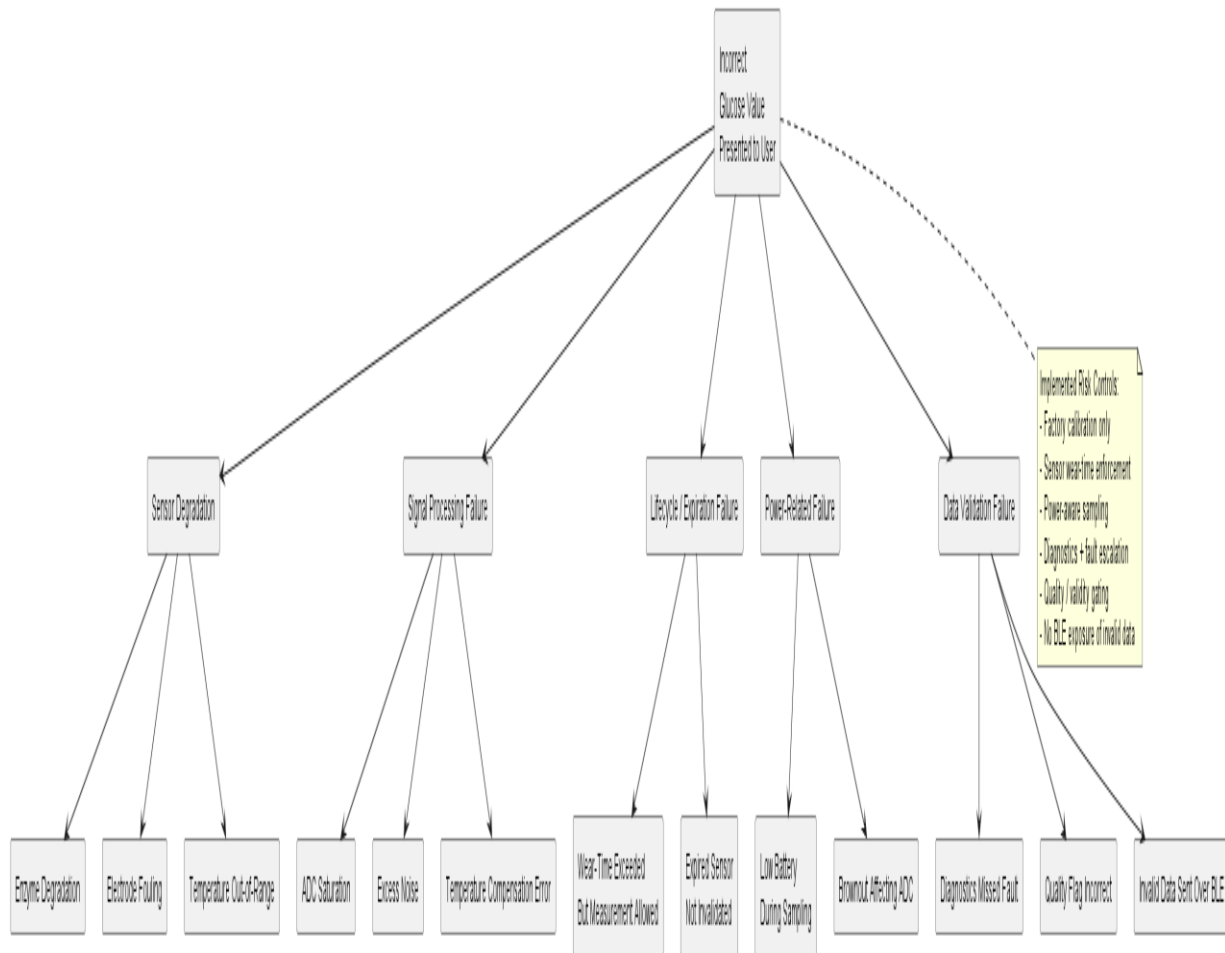
5.3 Alarm Path



Alarm Path

5.4 Fault Tree Analysis (ISO 14971)

Fault Tree Analysis - Incorrect Glucose Value



FTA